

A Machine Learning Model for 3D Face Identification in Clinical Admission

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Abstract: Accurate and efficient patient identification remains a major challenge in healthcare admission processes, where traditional manual procedures often lead to delays and operational inefficiencies. This work presents a machine learning pipeline for 3D face Identification, leveraging both Euclidean distances and geometric descriptors extracted from RGB-D images to enable secure and contactless patient registration. Data were collected from the Bosphorus public database and an internal dataset acquired with a RealSense sensor, ensuring diversity in acquisition devices and subject variability. After preprocessing and feature selection using the Fisher score, multiple classifiers such as Support Vector Machine (SVM), were evaluated. Among them the SVM with radial basis function (RBF) kernel achieved the highest recognition rate (0.98) and a zero false acceptance rate, confirming its robustness and suitability for security-critical clinical scenarios. The results demonstrate that the proposed approach attains state-of-the-art performance in controlled conditions and supports the integration of 3D face Identification systems in healthcare workflows to enhance both security and procedural efficiency.

Keywords: 3D face identification; machine learning; biometric identification; healthcare admission; geometric descriptors; Euclidean distances; SVM; security.

1. Introduction

Accurate and efficient patient registration within healthcare systems constitutes one of the most significant operational challenges. In outpatient or hospital admission settings, preliminary data collection phases require on average at least five minutes per patient, thereby exacerbating queues and reinforcing perceptions of inefficiency [1]. Waiting time represents a critical factor in the occurrence of queue abandonment, with early exit rates exceeding 5% in emergency departments [2]. When multiplied by high daily patient volumes, these timeframes create congestion at registration points and compromise the perceived quality of service. Although digital solutions can speed up data entry, they often do not address the specific usability barriers faced by older adults or people with cognitive and motor impairments, making these systems less accessible to vulnerable populations [3]. In addition to user dissatisfaction, manual registration procedures and traditional administrative management generate high labour costs and impose a substantial organisational burden on healthcare staff responsible for admission phases [4]. These factors negatively affect the overall efficiency of the system and consume operational resources that could be allocated to activities with greater clinical value. Recent evidence indicates that biometric authentication systems achieve Health Insurance Portability and Accountability Act (HIPAA) compliance scores that are higher than password-based solutions, while also providing more complete and traceable access log records [5].

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In this context, facial recognition emerges as a high-potential technology for optimising patient reception and registration procedures. The effectiveness of this tool has already been documented within the hospitality sector, where the automation of two-dimensional (2D) identity checks contributes to faster check-in operations and enhances perceived service quality [6]. Few studies have systematically examined facial recognition for patient identification at hospital admission, mainly due to stringent data security and biometric reliability requirements. These challenges limit adoption in clinical settings. However, three-dimensional (3D) facial recognition systems may offer greater accuracy and robustness than 2D.

From a technical perspective, 3D facial recognition systems rely on the processing of geometric information that enables detailed morphological reconstruction of the face. This modelling capability ensures greater accuracy in identifying distinctive features. Such characteristics render 3D-dimensional techniques particularly suitable for automated admission contexts, where unambiguous matching between the stored digital identity and the physical individual entering the facility must be ensured.

Numerous studies have investigated the use of machine learning in 3D facial recognition. Approaches based on facial landmarks, global information and local geometric descriptors, or deep neural networks have been developed [7]. Realistic datasets, such as Bosphorus and Lock3DFace, have been designed to simulate clinical scenarios and have demonstrated the effectiveness of these methods. However, challenges remain concerning the physiological variability of the face, including ageing and facial expressions. Further issues include occlusion caused by medical devices and the suboptimal quality of consumer-grade sensors [8]. To address these obstacles, methods such as Rotational Projection Statistics (RoPS) [9] and adaptive segmentation have been proposed. These techniques aim to increase recognition robustness under unfavourable conditions. Notably, Thatere et al. [10] have recently demonstrated that approaches based on handcrafted features and Support Vector Machines (SVM) can achieve high accuracy even in the presence of severe occlusions, confirming the effectiveness of traditional machine learning solutions in clinical and real-world scenarios.

This work describes a smart clinic model in which each patient is preregistered through secure facial templates uniquely associated with the individual's Electronic Health Record (EHR). A contactless facial scan is performed at the start of hospital admission. The patient's personal and medical information is quickly retrieved, and the need to fill out forms by hand is eliminated. The implemented algorithm is based on a Face Identification system. This involves a multiclass classification task designed to determine which subject, among those preregistered in the database, is physically present at the access point. This procedure aims to ensure that the digital identity corresponds unambiguously to the real individual, thereby reducing check-in times and matching errors. The key contributions of this work include the development of a machine learning (ML) system that leverages depth information for subject recognition, specifically targeting registered patients. The system employs Euclidean distances and geometric features to perform the recognition task.

The structure of the paper is organized as follows: the experimental approach adopted for the development of the proposed system is in the *Methods and Materials* section (2); The following sections are *Results* (3), *Discussion* (4) and *Conclusions* (5).

2. Materials and Methods

The methodologies adopted that led to the implementation of the final strategy, illustrated in Figure 1, will be discussed below.

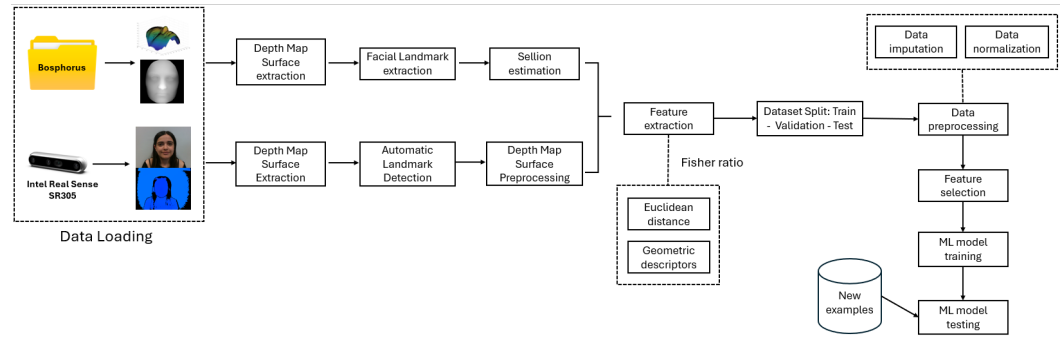


Figure 1. Pipeline of the project.

2.1. Data Structure

To establish a comprehensive and diverse dataset for facial Identification research, data was collected from two distinct sources:

Bosphorus 3D Face Database. The Bosphorus 3D Face Database is a publicly available repository containing 3D facial scans and textures of 105 individuals acquired using structured light sensors. The database includes multiple pose variations, realistic occlusions, and expressions based on Facial Action Coding System (FACS) Action Units and six basic emotions, with aligned 2D and 3D facial landmarks and surface coordinates available [11]. For this study, 27 subjects were selected from the database based on the availability of five specific pose configurations. These poses were chosen for their clinical relevance and reproducibility in real-world patient check-in scenarios: two neutral poses (N_N); two smiling poses (LFAU_12 and LFAU_12LW); one HAPPY pose representing a more dynamic and pronounced emotional expression.

Internal Dataset. Other 8 additional subjects were recruited and recorded in a controlled laboratory environment at Politecnico di Torino. Data acquisition was performed using an Intel RealSense SR305 camera, a coded light depth sensor optimized for close-range applications (0.2–1.5 meters). The SR305 employs infrared structured light projection to generate depth information, simultaneously capturing RGB and depth streams at 640×480 pixel resolution and 30 frames per second. The acquisition protocol was standardized to ensure data consistency. Participants were seated in an upright position against a uniform background under constant artificial lighting conditions. The camera was positioned at face level, maintaining a 30-centimeter distance from the subject. Each participant replicated the same five poses extracted from the Bosphorus database, ensuring consistency across datasets. Raw data was captured in .bag format and subsequently processed using RealSense Viewer software to extract individual depth frames in .raw format and corresponding RGB snapshots in .png format. Pixel-to-pixel correspondence between 2D (RGB) and 3D (depth) modalities was preserved through the alignment performed by *Align_RGBD* application.

2.2. Data Preparation

2.2.1. Data Extraction

Following subject selection from the Bosphorus 3D Face Database, the initial preprocessing phase involved reconstructing 3D facial surface depth in real spatial coordinates X , Y , and Z [11]. The subsequent step involved extraction and organization of 3D landmarks. The Bosphorus database provides .lm3 format files for each subject, containing 3D coordinates of standardized facial landmarks manually annotated by the database authors. The landmarks provided by the Bosphorus database include key anatomical points such as ocular, nasal, labial, and mental landmarks. The landmarks provided by the Bosphorus

do not include the *Sellion*, a landmark that has been widely used for calculating Euclidean distances in face identification tasks [12–14]. The *Sellion* is defined as the most posterior point of the frontonasal soft tissue contour on the midline of the nasal root base [15]. The estimation is based on the analysis of the Gaussian curvature of the facial surface. The following steps are involved in this process:

1. **Midline extraction:** the central column of the X-coordinate matrix, corresponding to the facial mid-sagittal line where the *Sellion* is located, is identified.
2. **Region of interest:** a bounding box is identified by leveraging known landmarks (i.e., the inner eyebrows and inner corners of the eyes), restricting the target area.
3. **Gaussian curvature analysis:** within the region of interest, the Gaussian curvature matrix (K) is analysed. This descriptor is employed to quantify the solid angle occupied by the beam of normals in an infinite patch, thereby providing a criterion for identifying pronounced concavity and convexity [16]. It has been determined that the point at which the value of K is close to 0 is referred to as the *Sellion* [14]. In particular, the negative value of K closest to zero was empirically determined, and it was found to correspond to the maximum local concavity. The coordinates (X , Y , Z) of the selected point are assigned as *Sellion*.

This approach is entirely automatic, repeatable and based on an objective geometric definition (Figure 2). The employment of Gaussian curvature facilitates the identification of the *Sellion*, even in the presence of individual morphological variability.

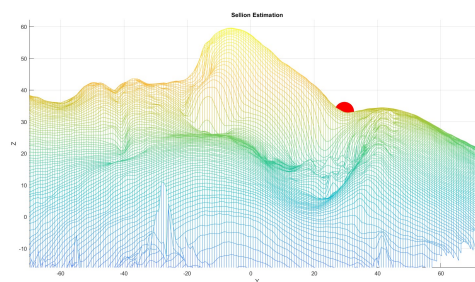


Figure 2. Sellion Estimation.

2.2.2. Data Acquisition

Subsequent to the collection of data, each sample is composed of two spatially aligned files: an RGB image (.png) and a depth map (.raw). The processing procedure consists of several complementary sequential steps as shown in Figure 3.

1. **Facial landmark extraction:** the RGB image is processed via the *MediaPipe* library to calculate the automatic facial mesh. The system has been developed to identify 468 standardised anatomical points on the face. The coordinates are returned normalised in the range $[0, 1]$ with respect to the image size. Subsequently, it was necessary to apply a denormalisation process that converted the values to pixel coordinates.
2. **Generation of facial masks:** a binary mask is generated from the complete *MediaPipe* mesh by tracing a closed contour around the detected landmarks.
3. **Depth map processing:** the depth map is loaded from the .raw file as a 16-bit matrix of distance values. Gaps and invalid data are automatically corrected by interpolation. The previously generated binary mask is then applied to segment and isolate only the depth values corresponding to the facial region.
4. **Selection and 3D projection of reference points:** in accordance with the protocol, the same reference points of the Bosphorus were selected. For each landmark, the 2D coordinates (x , y) are utilised to retrieve the depth value (z) from the segmented

matrix. The triplet (x, y, z) is then deprojected into the three-dimensional space using the camera's intrinsic parameters.

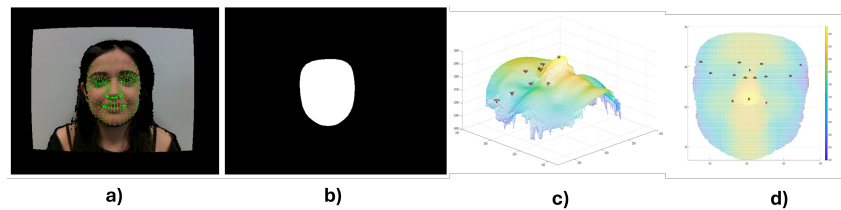


Figure 3. (a) Facial landmarks from MediaPipe, (b) Binary face mask from landmarks, (c) 3D side view of Z surface with landmarks, (d) Top view of Z surface with color map and landmarks.

2.3. Feature Extraction

Features were extracted from the Z surfaces of the subjects and 3D landmark coordinates, specifically selecting Euclidean distances and geometric descriptors as comparison parameters between images. [13].

2.3.1. Calculation of Euclidean Distances

From a review of the literature, the most frequently adopted Euclidean distances for face identification were identified. Specifically, following the study by Dagnes et al. [12], the 52 Euclidean were selected. The distances were chosen to comprehensively cover the facial surface, enhancing the descriptiveness level.

2.3.2. Geometric Descriptors

Based on a literature review, the most useful primary geometric descriptors for face identification were selected: S (shape index) and C (curvature) [16,17]. Additionally, based on the paper by [15], 9 derived descriptors were calculated, specifically those recognised as the most effective for landmark identification in 3D facial analysis. Geometric derived descriptors were obtained by applying standard mathematical functions to the primary descriptors. The 9 derived descriptors include $mean_g$, $median_g$, sin_g , $atan_g$, $median_H$, $atan_e$, sin_e , $atan_G$, and ln_E . Together with the primary descriptors S and C , these form the full set of geometric descriptors used for 3D facial characterization. Each descriptor is a matrix of surface point values, later summarized into histograms (bins) for compact representation. [15,17]. The Sturges rule was adopted to determine the optimal number of bins for all geometric descriptors except the shape index S . Applying Sturges rule to the dataset, a total of 15 bins for all geometric descriptors was determined. However, for the shape index, a different strategy was adopted: 9 bins with predefined intervals in the range $[-1, +1]$, according to the subdivision proposed by Koenderink and van Doorn [16]. Following the bin division of the geometric descriptors, a set of 159 geometric measures was obtained.

2.4. Dataset Structure

After calculating the Euclidean distances and geometric descriptors, a complete dataset with dimensions of 175×211 was generated. The 175 observations were derived from 35 subjects in 5 poses, and the 211 features included 159 geometric descriptors and 52 Euclidean distances. The class column was then added, assigning a unique label to each subject. Thirty-three subjects were labelled from 1 to 33, while two subjects were considered unknown and assigned class 0. The dataset of the 33 registered subjects was split into training, validation and test sets in the proportions 60%, 20%, and 20% respectively. After this split, the two unknown subjects (with all their poses) were added exclusively to the test set as class 0, enabling an unbiased evaluation of the system's open-set recognition

capability. The final dataset therefore consists of 165 samples from the registered subjects and 10 samples from the unknown subjects, for a total of 175×212 .

2.4.1. Data Cleaning and Data Normalisation

Before proceeding with the analysis, the dataset was inspected to identify any issues related to missing values. These missing values were due to specific landmarks not being detected in some RGB-D images. An imputation strategy was implemented to estimate the missing features. For the training set, the mean of the available values for each subject was calculated to impute missing values. For the validation and test sets, an automated procedure was applied where the overall mean for each feature was calculated within the training set and subsequently used to impute missing values in the test set. This approach ensures a systematic method of handling cases of missing data. To avoid bias resulting from different measurement scales, the dataset was normalised using z-score standardisation, a statistical technique that transforms the data to have a mean of zero and a standard deviation of one.

2.4.2. Feature Selection

Feature selection was performed using the ratio score, also known as the Fisher score, a statistical measure based on the ratio between inter-class variance and intra-class variance [18]. A high ratio score value indicates a strong ability of the feature to separate different classes, making it particularly useful for classification tasks. The score was calculated separately for Euclidean distances and geometrical descriptors. The 12 Euclidean distances with the highest significant values were selected.

The selection criterion was systematically applied to all bins of the computed geometric descriptors. Of these, the bins of the shape index and the derived descriptor *atan_G* which showed particularly high significance levels, were selected. Specifically, 18 features are chosen for further analysis. Figure 4 illustrates the Euclidean distances selected and the mesh visualisations of these two descriptors.

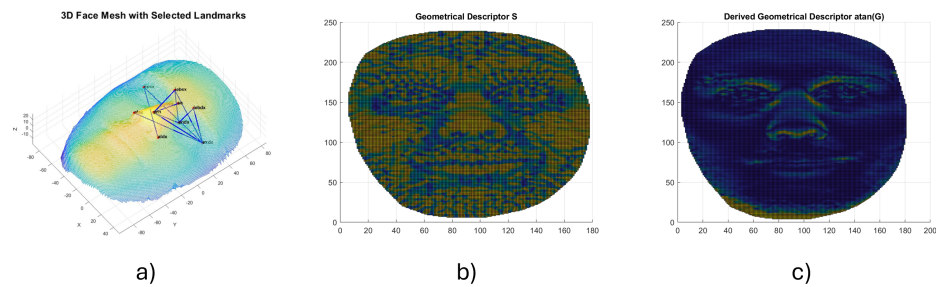


Figure 4. a) Euclidean distances, b) Shape Index, c) *atan(G)*.

2.5. Evaluation of Classifiers and Decision Thresholds

Several ML algorithms used in similar studies were tested and evaluated. The multi-class classifiers employed in this study were Support Vector Machine (SVM), K Nearest Neighbour (KNN), Random Forest (RF), and a Multi-Layer Perceptron (MLP) neural network. The selection of parameters for the various classifiers was made empirically, following extensive testing and performance evaluations, with the objective of optimising the recognition rate on the validation set.

A customised decision threshold was introduced for each model to handle the possibility that an individual does not belong to any of the faces in the gallery.

- SVM classifier was implemented with a radial basis function (RBF) kernel, a regularisation parameter $C=100$, decision threshold as 0.09 and a one-vs-one approach for multi-class problem.

- KNN used parameter k set to 2, following an empirical evaluation of neighbouring values to minimise the classification error. The acceptance threshold was set at 0.8.
- RF classifier was configured with 100 trees, a maximum depth of 10, and standard splitting parameters. The decision threshold was set at 0.09, in accordance with a conservative strategy aimed at preventing any unknown patient from being classified as registered, despite the consequent increase in the false non-match rate.
- MLP neural network was implemented with a single hidden layer of 100 neurons, trained for a maximum of 1000 epochs and acceptance threshold as 0.7.

Thresholds were empirically selected for each classifier by analysing the distribution of maximum prediction scores on the validation set, choosing the value that best minimised false positives.

The performance of the classifiers was evaluated using the following metrics [19]:

- **Recognition Rate (RR):** the overall accuracy of the classifier.

$$RR = \frac{\text{Correctly classified in total}}{\text{Total samples}} \quad (1)$$

- **Genuine Acceptance Rate (GAR):** the ratio of correct identifications when the input face images are indeed from the authenticated user.

$$GAR = \frac{\text{Correctly classified Registered Patients}}{\text{Total Registered Patients}} \quad (2)$$

- **False Acceptance Rate (FAR):** a measure of the likelihood that the system will incorrectly identify an individual as someone else. This is a significant measure of the security effectiveness of the system.

$$FAR = \frac{\text{Unknown Patients erroneously classified as Registered}}{\text{Total Unknown Patients}} \quad (3)$$

- **False Reject Rate (FRR):** the likelihood that the system incorrectly rejects a match. FRR is a crucial measure of the usability of the system.

$$FRR = \frac{\text{Registered Patients not recognised}}{\text{Total Registered Patients}} \quad (4)$$

2.6. Visualisation of the prototype system in Unity

The demonstrative prototype for interactive facial biometric visualisation is developed using Unity (*version 2022.3.6f1*), with logic components developed in C#. It is configured as an immersive virtual platform designed for the presentation and analysis of geometric relationships between facial landmarks. Although the application does not operate in real time, it was conceived for demonstrative and educational purposes.

2.6.1. System Architecture

The virtual environment consists of a 3D facial model developed and exported from Blender. This model includes a mesh with visible landmarks and the Euclidean distances. The user interface (UI), organised within a Canvas GameObject, is divided into functional panels that allow model manipulation, biometric data interaction, and simulation of the facial scanning process.

2.6.2. Operational Flow

The interaction flow follows a logical and user-friendly sequence. At application launch, the 3D facial mesh is immediately displayed, allowing the user to begin a preliminary biometric analysis through the dedicated interface panel. The model can be interactively manipulated: users may rotate and zoom the mesh for detailed inspection, as

well as adjust its transparency and visibility to focus on specific facial regions. For biometric analysis, the system enables dynamic visualization of distances between anatomical landmarks. Users can view the complete network of landmark connections or select individual segments to examine specific measurements, with all distances clearly annotated directly on the model. A biometric scan simulation is available, during which a scanning line moves across the mesh to visually represent the matching process. Progress is continuously updated, and upon completion, a panel displays the patient's information along with a direct link to their digital medical records, streamlining access to essential data. A central coordinator manages the reset functionality, allowing users to restore the application to its initial state.

3. Results

The following section summarizes the main results of the face identification system tested in a clinical acceptance scenario. In Figure 5, the recognition rates obtained by each classifier (SVM, KNN, RF, MLP) on the training, validation, and test sets are shown. The SVM achieved the highest recognition rate on all sets, followed by KNN and MLP, while RF showed lower performance, especially on the test set.

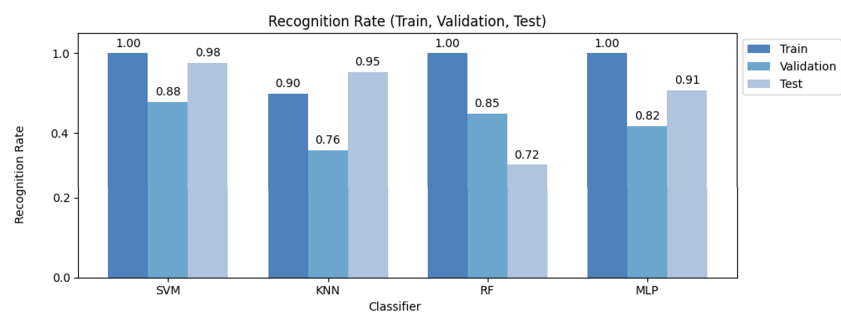


Figure 5. Recognition rate for the training, validation, and test sets for the different classifiers.

Table 1 reports, for each classifier, the number of *Registered Patients* (patients previously enrolled in the system) and *Unknown Patients* (not enrolled), indicating how many were correctly or incorrectly classified. Table 2 summarizes the main biometric evaluation metrics RR, GAR, FAR and FRR for the SVM classifier.

Table 1. Results for all classifiers.

Classifier	Subject	No of Samples	Correctly Classified	Incorrectly Classified
SVM	Registered	33	32	1
	Unknown	10	10	0
KNN	Registered	33	31	2
	Unknown	10	10	0
RF	Registered	33	31	2
	Unknown	10	0	10
MLP	Registered	33	29	4
	Unknown	10	10	0

Table 2. RR, GAR, FAR, FRR of SVM classifier.

RR	GAR	FAR	FRR
0.98	0.97	0.00	0.03

4. Discussion

The analysis of the results highlights that the SVM classifier achieves the highest recognition rates across all sets, with 1.00 on the training set, 0.88 on the validation set, and 0.98 on the test set. The MLP classifier also demonstrates good performance, especially on the test set (0.91), but remains slightly below SVM. In contrast, the KNN and RF classifiers show lower recognition rates, particularly on the validation and test sets, with KNN reaching 0.76 and 0.95, and RF 0.72 and 0.85, respectively. These results confirm the superior generalizability of SVM in this context, while KNN and RF are more sensitive to variability in the data. In addition to the RR, a comprehensive evaluation of the SVM classifier was performed using standard biometric metrics, as summarised in Table 2. The GAR was 0.97, meaning that almost all registered patients were correctly identified. The FAR was 0%, confirming that no unknown patient was incorrectly accepted as registered — an essential requirement for security-critical applications. The FRR was 0.03, reflecting a small proportion of genuine users who were not recognised by the system and would require manual verification. These results highlight the system's ability to maintain a high level of security (by minimising false acceptances) while preserving usability for legitimate users (with a high GAR and a low FRR), in line with the recommendations and evaluation practices highlighted in literature [19]. Placing these results in the context of existing literature allows for a better assessment of the effectiveness of the proposed methods. Faruque et al. [20] conducted face recognition experiments on the Olivetti Research Laboratory (ORL) database, which consists of 400 grayscale images from 40 individuals (10 images per subject), acquired under controlled conditions with variations in expression and minor pose changes. In their protocol classification was performed using SVM with linear, polynomial, and radial basis function (RBF) kernels. Their best result was achieved with the RBF-SVM, reaching a recognition rate of 97.25% when using 30 principal components as features. In this study, two controlled datasets were used: Bosphorus and the internal one. Using an SVM with RBF kernel, the system achieved a recognition rate of 98% on the test set, fully in line with the results reported by Faruque et al. [20] on the ORL dataset. This comparison confirms that SVM with RBF kernel, combined with appropriate feature extraction, achieves state-of-the-art performance for face recognition in classical ML pipelines, even when applied to different controlled datasets and acquisition devices.

Table 3. Recognition rates achieved by RBF-SVM on the ORL dataset [20] and on the Bosphorus plus internal dataset used in this study.

Dataset	ML Model	Recognition Rate (RR)
ORL	RBF-SVM	97.25%
Bosphorus + internal	RBF-SVM	98%

The primary contribution of this work lies in the implementation and evaluation of a robust face Identification pipeline based on classical ML, validated on datasets with different acquisition devices and subject variability, and in the integration of a security-oriented thresholding strategy. Despite the encouraging outcomes observed with the proposed methodology, it is acknowledged that certain limitations remain that necessitate further refinement to be fully realised. The primary constraint relates to the limited size of the datasets and the employment of controlled acquisition conditions. Furthermore, the conservative thresholding approach, while effective in preventing impostor acceptance, increases the FRR, potentially reducing usability in real-world scenarios. It is recommended that future research should extend the dataset to include a greater number of subjects and acquisition conditions, real-time, as well as the exploration of hybrid approaches combining classical and deep learning methods to further improve both security and usability.

5. Conclusions

The challenge of reliable and efficient patient identification in clinical admission settings is addressed through the development of a ML pipeline that combines Euclidean distances and geometric descriptors derived from 3D facial data. The proposed approach demonstrates high RR and robust security, given a null FAR. The system's performance is consistent with state-of-the-art results reported in the literature, confirming the effectiveness of classical ML methods in controlled acquisition scenarios. Notable contributions include the integration of a security-oriented thresholding strategy and validation on heterogeneous datasets. Future research should focus on expanding the dataset, incorporating more diverse acquisition scenarios, and exploring hybrid or deep learning solutions to further enhance both usability and security in real-world healthcare environments.

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